WORKED SOLUTION

STEP III 1999, Question 2

Pure • Counting Real Roots

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q2

The key idea

In part (i) the derivative factorises as a perfect square times e^x, so the function never decreases, and it is in fact strictly increasing. In part (ii) the derivative of the new function vanishes exactly when $(1+x^2)e^x = k$, which is the equation from part (i). That tells us the new function has a single turning point, a minimum. After that, everything depends on what happens as $x \to -\infty$, where $\tan^{-1} x \to -\frac{\pi}{2}$. This is where the number $2/\pi$ comes from.

Solution

Part (i)

By the product rule,

$$f'(x) = 2x\,e^x + (1+x^2)e^x = e^x\left(x^2+2x+1\right) = e^x(1+x)^2.$$

Since $e^x > 0$ and $(1+x)^2 \geq 0$, we have $f'(x) \geq 0$ for all x, as required. Equality holds only at $x = -1$.

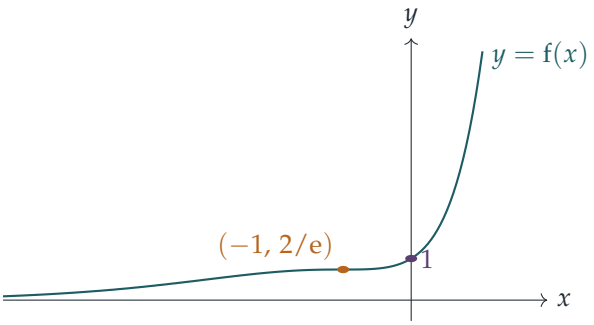
Ask yourself $f'(-1) = 0$, so is f really *strictly* increasing?

Yes. If $a < b$, then $f(b) - f(a) = \int_a^b f'(x)\,dx$, and the integrand is continuous, non-negative and zero at no more than one point of $[a, b]$, so the integral is strictly positive. A single stationary point can't stop a function from increasing. Here $x = -1$ is a stationary point of inflection: f' is positive on both sides of it.

For the sketch, we need:

- $f(x) > 0$ for all x, since both factors are positive;
- $f(0) = 1$, and the point of inflection is $(-1, 2/e) \approx (-1, 0.74)$;
- as $x \to \infty$, $f(x) \to \infty$;

- as $x \to -\infty$, $\mathrm{f}(x) \to 0$, because e^x decays faster than any polynomial grows ($x^2\mathrm{e}^x \to 0$). So the negative x-axis is an asymptote.



Now f is continuous and strictly increasing, with $\mathrm{f}(x) \to 0$ as $x \to -\infty$ and $\mathrm{f}(x) \to \infty$ as $x \to \infty$. So it takes every value in $(0, \infty)$ (intermediate value theorem), and it takes each one exactly once (strictly increasing). It never takes a value ≤ 0, since $\mathrm{f} > 0$. Therefore $(1 + x^2)\mathrm{e}^x = k$ has exactly one real root when $k > 0$ and none when $k \leq 0$. ■

Part (ii)

Let

$$\mathrm{g}(x) = \mathrm{e}^x - 1 - k\tan^{-1} x.$$

Straight away, $\mathrm{g}(0) = 1 - 1 - 0 = 0$, so $x = 0$ is always a root. Next,

$$\mathrm{g}'(x) = \mathrm{e}^x - \frac{k}{1+x^2} = \frac{(1+x^2)\mathrm{e}^x - k}{1+x^2} = \frac{\mathrm{f}(x) - k}{1+x^2}.$$

In both cases $k > 0$, so by part (i) there is exactly one number a with $\mathrm{f}(a) = k$. Because f is strictly increasing, $\mathrm{f}(x) < k$ for $x < a$ and $\mathrm{f}(x) > k$ for $x > a$. Hence

$$\mathrm{g}'(x) < 0 \text{ for } x < a, \qquad \mathrm{g}'(x) > 0 \text{ for } x > a.$$

So g is strictly decreasing on $(-\infty, a]$ and strictly increasing on $[a, \infty)$, with a single minimum at $x = a$. In particular g has at most one root on each side of a, so at most two roots in total.

Where is a? In both cases $k < 1 = \mathrm{f}(0)$, so $\mathrm{f}(a) = k < \mathrm{f}(0)$, which gives $a < 0$. Then $x = 0$ lies in the increasing stretch $[a, \infty)$. It is the only root there, and

$$\mathrm{g}(a) < \mathrm{g}(0) = 0.$$

What remains is to decide whether g also has a root in $(-\infty, a)$.

Behaviour as $x \to -\infty$. Since $\mathrm{e}^x \to 0$ and $\tan^{-1} x \to -\frac{\pi}{2}$,

$$\mathrm{g}(x) \to -1 + \frac{k\pi}{2} = \frac{\pi}{2}\left(k - \frac{2}{\pi}\right) \quad \text{as } x \to -\infty.$$

Call this limit L. Because g is strictly decreasing on $(-\infty, a]$, every value of g there is strictly less than L. To see this, take $x \leq a$ and any $x' < x$. Every $y < x'$ has $\mathrm{g}(y) > \mathrm{g}(x')$, so letting $y \to -\infty$ gives $L \geq \mathrm{g}(x')$. Hence $\mathrm{g}(x) < \mathrm{g}(x') \leq L$.

(a) $0 < k \leq 2/\pi$**.** Here $L \leq 0$, so $\mathrm{g}(x) < L \leq 0$ for every $x \leq a$. There is no root to the left of a, and the equation has **exactly one real root**, namely $x = 0$.

(b) $2/\pi < k < 1$**.** Here $L > 0$, so $\mathrm{g}(x) > 0$ for all sufficiently negative x, while $\mathrm{g}(a) < 0$. By the intermediate value theorem there is a root in $(-\infty, a)$, and it is the only one there because g is strictly decreasing. Together with $x = 0$, the equation has **exactly two real roots**: one negative root (less than a) and $x = 0$.

? **Ask yourself** Why does the boundary case $k = 2/\pi$ belong with (a)?

When $k = 2/\pi$, the left-hand limit is exactly $L = 0$. The decreasing branch approaches 0 from below but never reaches it, so no second root appears. Strict monotonicity is what settles this.

Checks

- Differentiating f numerically at random points agrees with $\mathrm{e}^x(1+x)^2$, and $\mathrm{f}(-1) = 2/\mathrm{e} \approx 0.736$.
- We counted sign changes of g on a fine grid over $[-200, 10]$. There is one root for $k = 0.05, 0.3, 0.5, 0.6$ and $2/\pi \approx 0.6366$, and two roots for $k = 0.65, 0.7, 0.8, 0.9, 0.99$. For $k = 0.65$ the negative root is far out (near $x \approx -31$), because $L = 0.65 \cdot \frac{\pi}{2} - 1 \approx 0.021$ is so small.
- The minimum point is $a \approx -1.81$ for $k = 0.7$, and $a \to 0^-$ as $k \to 1^-$. This fits: at $k = 1$, $\mathrm{g}'(0) = 0$ and $x = 0$ becomes a double root.

Common slips

- Forgetting that $x = 0$ is always a root, and so counting only the roots that come from the sign change at $-\infty$.
- Taking $\tan^{-1} x \to -\infty$ as $x \to -\infty$. It is bounded, tending to $-\frac{\pi}{2}$.
- Concluding from $\mathrm{f}'(-1) = 0$ that f has a turning point at $x = -1$.

Beyond STEP

The argument here is a model of a standard real-analysis technique: a continuous function that is strictly monotonic on an interval has at most one zero there, and the intermediate value theorem (applied with limits at infinity) tells us when it has

one. Counting roots by locating the turning points of a function, and then checking signs at the turning points and at the ends, is the idea behind Rolle's theorem and its consequences. Abbott's *Understanding Analysis* develops continuity, the intermediate value theorem and the mean value theorem carefully.

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